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[Portfolio Website](#)

[LinkedIn](#)

[GitHub](#)

Summary

Software Engineer | AI, Analytics & Scalable Systems:

Computer Engineer specializing in the intersection of scalable software architecture and complex AI/Machine Learning models. Combines a deep foundation spanning from embedded systems and IoT to cloud-native APIs. Proven track record of bridging hardware-software co-design with production-ready data pipelines using Java (Spring Boot), Go, and robust SQL databases. Accustomed to fast-paced, collaborative project sprints and fully primed to bring a "builder" mindset to cross-industry engineering and digital transformation projects.

Skills

- **Software Engineering & APIs:** Java (Spring Boot), Go, Python, C/C++, API Design (REST, Open API, GraphQL)
- **AI, Analytics & Math:** Machine Learning Inference, Optimization Algorithms, Linear Algebra, Data Mining, Statistics, Neural Networks, Reinforcement Learning.
- **Hardware & Embedded:** Internet of Things (IoT), Assembly, Hardware-Software Co-Design, Microprocessors, Embedded Systems.
- **Architecture & Databases:** SQL (PostgreSQL, SQL Server), Microservices (Eureka), Kafka, Event-Driven Architecture.
- **Infrastructure:** Docker, Git, CI/CD pipelines (GitHub actions, Jenkins)

Education and training

Student, Further Vocational Qualification in ICT | Grit:lab (Mariehamn, Åland) Sep 2024 – Aug 2026

- **Focus:** Intensive, project-based collaborative sprints focusing on scalable software (Go, Java/Spring Boot) and Machine Learning algorithms (Python, PyTorch, scikit learn, pandas).
- **Status:** In final stages, fully primed to transition into a fast-paced, full-time professional engineering role.

Bachelor of Science, Computer Engineering | University of Kashan (Kashan, Iran)

- **Core Hardware & Systems Coursework:** Internet of Things (IoT), Machine Language & Microprocessors, Hardware-Software Co-Design, Computer Networks, Operating Systems.
- **Core Data & AI Coursework:** Artificial Intelligence, optimization, Data Mining, Algorithm Design & Analysis, Database Implementation.
- **Teaching Experience:** Teaching Assistant - Data Structures (fostering deep understanding of scalable code, memory management, and computational efficiency).

Projects

buy-01 (E-commerce Microservices) | [GitHub](#)

- Architected a highly scalable, cloud-ready Java (Spring Boot) backend, managing complex transactional workflows and user configurations.
- Engineered event-driven asynchronous communication using Kafka and implemented robust NoSQL database interactions to ensure high-performance data management.
- Designed RESTful APIs and integrated CI/CD pipelines to maintain secure, enterprise-grade deployment standards.

Evacuation Planning Optimization | [GitHub](#)

- Developed a complex multi-objective optimization model using the NSGA-II genetic algorithm in Python.
- Demonstrated the ability to handle high-dimensional logic, tune hyperparameters, and prioritize computational efficiency in resource-constrained environments.

Matrix Factorization | [GitHub](#)

- Developed a recommendation engine using Singular Value Decomposition (SVD) and Bayesian Probabilistic matrix factorization (PMF) to handle highly sparse datasets (MovieLens).

Spectral Learning | [GitHub](#)

- Performed data decomposition on the Wine Quality dataset to identify hidden patterns and latent features. Implemented two different mathematical approaches, SVD and PCA

Languages

Persian: Native | English: Advanced | Swedish: Beginner